covid_19

January 11, 2025

1 Import libraries

[1]:
```
import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

2 Data exploration

[2]:
```
data = pd.read_csv('country_wise_latest.csv')
```

[3]:
```
pd.concat([data.head(),data.tail()])
```

[3]:

	Country/Region	Confirmed	Deaths	Recovered	Active	New cases	New deaths	New recovered	Deaths / 100 Cases	Recovered / 100 Cases
0	Afghanistan	36263	1269	25198	9796	106	10	18	3.50	69.49
1	Albania	4880	144	2745	1991	117	6	63	2.95	56.25
2	Algeria	27973	1163	18837	7973	616	8	749	4.16	67.34
3	Andorra	907	52	803	52	10	0	0	5.73	88.53
4	Angola	950	41	242	667	18	1	0	4.32	25.47
182	West Bank and Gaza	10621	78	3752	6791	152	2	0	0.73	35.33
183	Western Sahara	10	1	8	1	0	0	0	10.00	80.00
184	Yemen	1691	483	833	375	10	4	36	28.56	49.26
185	Zambia	4552	140	2815	1597	71	1	465	3.08	61.84
186	Zimbabwe	2704	36	542	2126	192	2	24	1.33	20.04

	Deaths / 100 Recovered	Confirmed last week	1 week change	\
0	5.04	35526	737	
1	5.25	4171	709	
2	6.17	23691	4282	
3	6.48	884	23	
4	16.94	749	201	
182	2.08	8916	1705	
183	12.50	10	0	
184	57.98	1619	72	
185	4.97	3326	1226	
186	6.64	1713	991	

	1 week % increase	WHO Region
0	2.07	Eastern Mediterranean
1	17.00	Europe
2	18.07	Africa
3	2.60	Europe
4	26.84	Africa
182	19.12	Eastern Mediterranean
183	0.00	Africa
184	4.45	Eastern Mediterranean
185	36.86	Africa
186	57.85	Africa

[4]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 187 entries, 0 to 186
Data columns (total 15 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Country/Region          187 non-null    object 
 1   Confirmed               187 non-null    int64  
 2   Deaths                  187 non-null    int64  
 3   Recovered               187 non-null    int64  
 4   Active                  187 non-null    int64  
 5   New cases               187 non-null    int64  
 6   New deaths              187 non-null    int64  
 7   New recovered           187 non-null    int64  
 8   Deaths / 100 Cases      187 non-null    float64
 9   Recovered / 100 Cases   187 non-null    float64
 10  Deaths / 100 Recovered  187 non-null    float64
 11  Confirmed last week     187 non-null    int64  
 12  1 week change           187 non-null    int64  
 13  1 week % increase       187 non-null    float64
 14  WHO Region              187 non-null    object 
dtypes: float64(4), int64(9), object(2)
```

```
memory usage: 22.0+ KB
```

[5]:
```
data.describe(include='all')
```

```
e:\programs\anaconda3\envs\my_env\lib\site-packages\pandas\core\nanops.py:1016:
RuntimeWarning: invalid value encountered in subtract
  sqr = _ensure_numeric((avg - values) ** 2)
```

[5]:

	Country/Region	Confirmed	Deaths	Recovered
count	187	1.870000e+02	187.000000	1.870000e+02
unique	187	NaN	NaN	NaN
top	Afghanistan	NaN	NaN	NaN
freq	1	NaN	NaN	NaN
mean	NaN	8.813094e+04	3497.518717	5.063148e+04
std	NaN	3.833187e+05	14100.002482	1.901882e+05
min	NaN	1.000000e+01	0.000000	0.000000e+00
25%	NaN	1.114000e+03	18.500000	6.265000e+02
50%	NaN	5.059000e+03	108.000000	2.815000e+03
75%	NaN	4.046050e+04	734.000000	2.260600e+04
max	NaN	4.290259e+06	148011.000000	1.846641e+06

	Active	New cases	New deaths	New recovered
count	1.870000e+02	187.000000	187.000000	187.000000
unique	NaN	NaN	NaN	NaN
top	NaN	NaN	NaN	NaN
freq	NaN	NaN	NaN	NaN
mean	3.400194e+04	1222.957219	28.957219	933.812834
std	2.133262e+05	5710.374790	120.037173	4197.719635
min	0.000000e+00	0.000000	0.000000	0.000000
25%	1.415000e+02	4.000000	0.000000	0.000000
50%	1.600000e+03	49.000000	1.000000	22.000000
75%	9.149000e+03	419.500000	6.000000	221.000000
max	2.816444e+06	56336.000000	1076.000000	33728.000000

	Deaths / 100 Cases	Recovered / 100 Cases	Deaths / 100 Recovered
count	187.000000	187.000000	187.00
unique	NaN	NaN	NaN
top	NaN	NaN	NaN
freq	NaN	NaN	NaN
mean	3.019519	64.820535	inf
std	3.454302	26.287694	NaN
min	0.000000	0.000000	0.00
25%	0.945000	48.770000	1.45
50%	2.150000	71.320000	3.62
75%	3.875000	86.885000	6.44
max	28.560000	100.000000	inf

	Confirmed last week	1 week change	1 week % increase	WHO Region

```
count          1.870000e+02     187.000000         187.000000        187
unique                  NaN            NaN                NaN          6
top                     NaN            NaN                NaN     Europe
freq                    NaN            NaN                NaN         56
mean           7.868248e+04    9448.459893          13.606203        NaN
std            3.382737e+05   47491.127684          24.509838        NaN
min            1.000000e+01     -47.000000          -3.840000        NaN
25%            1.051500e+03      49.000000           2.775000        NaN
50%            5.020000e+03     432.000000           6.890000        NaN
75%            3.708050e+04    3172.000000          16.855000        NaN
max            3.834677e+06  455582.000000         226.320000        NaN
```

3 Data Check conflict

[6]:
```
#nothing here
```

4 Drop unecessary Features

[7]:
```
#nothing here
```

5 Data Types

[8]:
```
data.columns
```

[8]:
```
Index(['Country/Region', 'Confirmed', 'Deaths', 'Recovered', 'Active',
       'New cases', 'New deaths', 'New recovered', 'Deaths / 100 Cases',
       'Recovered / 100 Cases', 'Deaths / 100 Recovered',
       'Confirmed last week', '1 week change', '1 week % increase',
       'WHO Region'],
      dtype='object')
```

[9]:
```
pd.DataFrame({"Unique": data.nunique() ,"Type":data.dtypes}).T
```

[9]:
```
        Country/Region Confirmed Deaths Recovered Active New cases New deaths  \
Unique             187       184    150       178    173       122         38
Type            object     int64  int64     int64  int64     int64      int64

        New recovered Deaths / 100 Cases Recovered / 100 Cases  \
Unique            103                145                   177
Type            int64            float64               float64

        Deaths / 100 Recovered Confirmed last week 1 week change  \
Unique                     155                 183           162
Type                   float64               int64         int64
```

```
        1 week % increase WHO Region
Unique                169          6
Type              float64     object
```

[10]:
```
num_features = ['Confirmed', 'Deaths', 'Recovered', 'Active','New cases', 'New␣
↪deaths', 'New recovered', 'Deaths / 100 Cases',
      'Recovered / 100 Cases', 'Deaths / 100 Recovered','Confirmed last week',␣
↪'1 week change', r'1 week % increase',]
cat_features = ['Country/Region', 'WHO Region']
```

[11]:
```
data[cat_features] = data[cat_features].astype('object')
data[num_features] = data[num_features].astype('float64')
```

[12]:
```
pd.DataFrame({"Unique": data.nunique() ,"Type":data.dtypes}).T
```

[12]:
```
        Country/Region Confirmed   Deaths Recovered   Active New cases  \
Unique             187       184      150       178      173       122
Type            object   float64  float64   float64  float64   float64

        New deaths New recovered Deaths / 100 Cases Recovered / 100 Cases  \
Unique          38           103                145                   177
Type       float64       float64            float64               float64

        Deaths / 100 Recovered Confirmed last week 1 week change  \
Unique                     155                183           162
Type                   float64            float64       float64

        1 week % increase WHO Region
Unique                169          6
Type              float64     object
```

6 Check Null values

[13]:
```
100 * data.isna().sum() / len(data)
```

[13]:
```
Country/Region            0.0
Confirmed                 0.0
Deaths                    0.0
Recovered                 0.0
Active                    0.0
New cases                 0.0
New deaths                0.0
New recovered             0.0
Deaths / 100 Cases        0.0
Recovered / 100 Cases     0.0
Deaths / 100 Recovered    0.0
Confirmed last week       0.0
```

```
1 week change              0.0
1 week % increase          0.0
WHO Region                 0.0
dtype: float64
```

7 Outliers

[14]:
```
len(num_features)
```

[14]:
```
13
```

[15]:
```
def IQR(feature):
    global data

    Q1, Q3 = data[feature].quantile(0.25), data[feature].quantile(0.75)
    IQR = Q3 - Q1
    min_v = Q1 - 1.5 * IQR
    max_v = Q3 + 1.5 * IQR

    outlier_min =  data[data[feature] < min_v][feature].values
    outlier_max =  data[data[feature] > max_v][feature].values

    data[feature] = data[feature].replace(outlier_min, min_v)
    data[feature] = data[feature].replace(outlier_max, max_v)
```

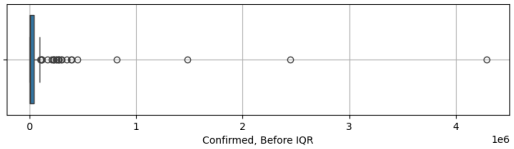
[16]:
```
for feature in num_features:
    plt.figure(figsize=(20,2))

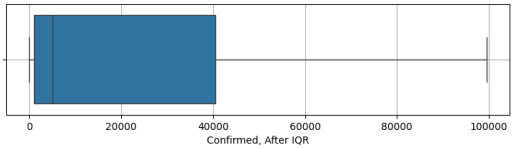
    plt.subplot(1, 2 , 1)
    sns.boxplot(data=data, x=feature)
    plt.xlabel(f'{feature}, Before IQR')
    plt.grid(True)

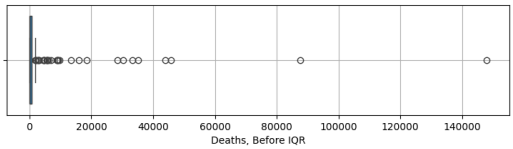
    IQR(feature)

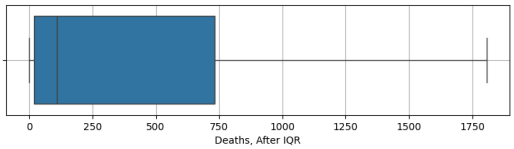
    plt.subplot(1, 2 , 2)
    sns.boxplot(data=data, x=feature)
    plt.xlabel(f'{feature}, After IQR')
    plt.grid(True)

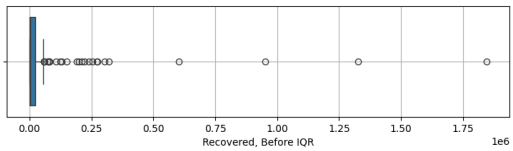
    plt.show()
```

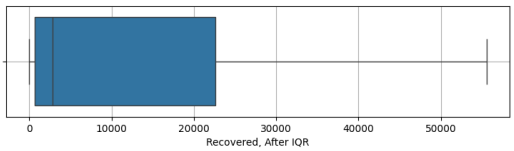


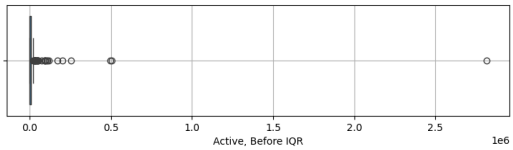


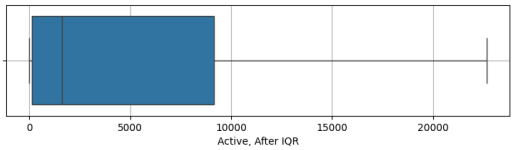

0
20000
40000
60000
80000
100000
120000
140000
Deaths, Before IQR

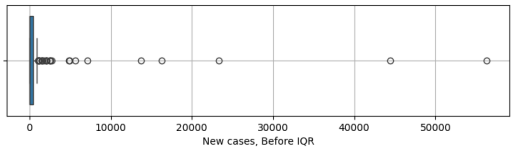

0
250
500
750
1000
1250
1500
1750
Deaths, After IQR

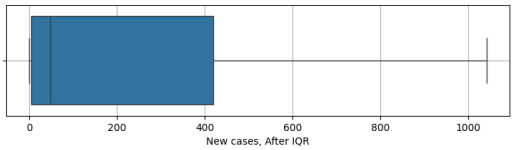

0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Recovered, Before IQR
1e6

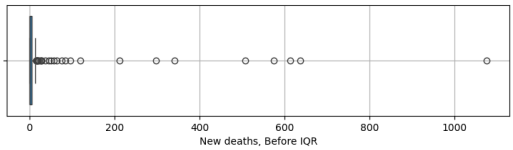

0
10000
20000
30000
40000
50000
Recovered, After IQR

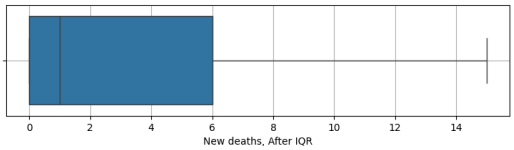

0.0
0.5
1.0
1.5
2.0
2.5
Active, Before IQR
1e6

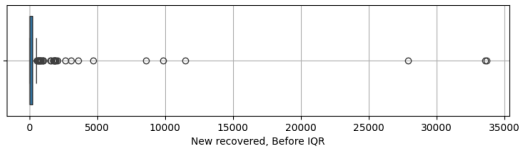

0
5000
10000
15000
20000
Active, After IQR

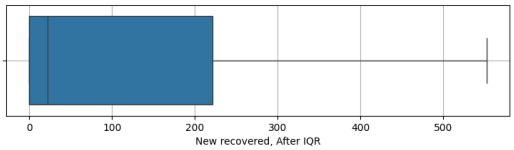

0
10000
20000
30000
40000
50000
New cases, Before IQR

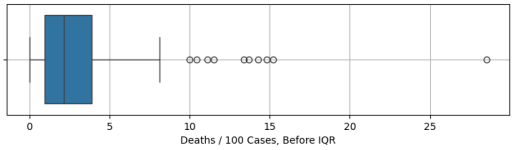

0
200
400
600
800
1000
New cases, After IQR

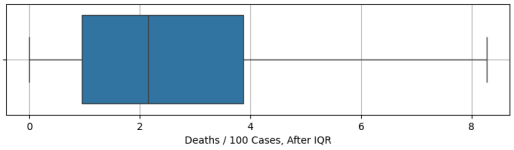

0
200
400
600
800
1000
New deaths, Before IQR

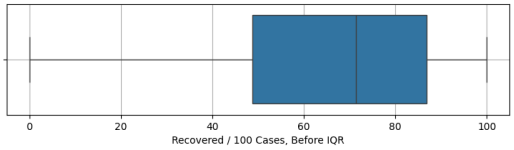

0
2
4
6
8
10
12
14
New deaths, After IQR

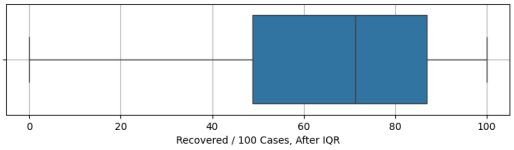

0
5000
10000
15000
20000
25000
30000
35000
New recovered, Before IQR

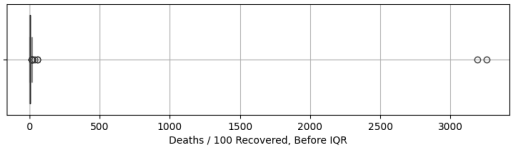

0
100
200
300
400
500
New recovered, After IQR

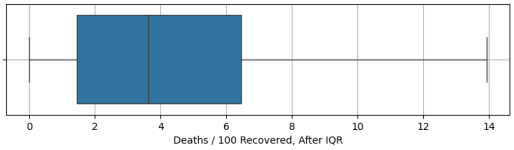

0
5
10
15
20
25
Deaths / 100 Cases, Before IQR

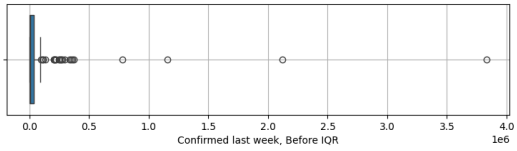

0
2
4
6
8
Deaths / 100 Cases, After IQR

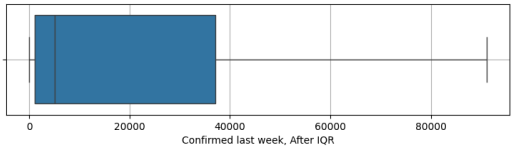

0
20
40
60
80
100
Recovered / 100 Cases, Before IQR


0
20
40
60
80
100
Recovered / 100 Cases, After IQR

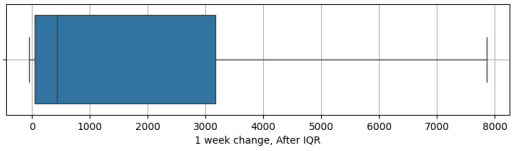

0
500
1000
1500
2000
2500
3000
Deaths / 100 Recovered, Before IQR

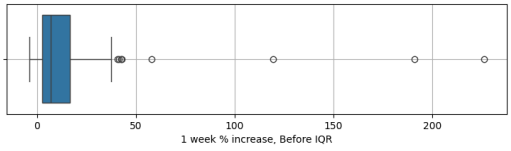

0
2
4
6
8
10
12
14
Deaths / 100 Recovered, After IQR

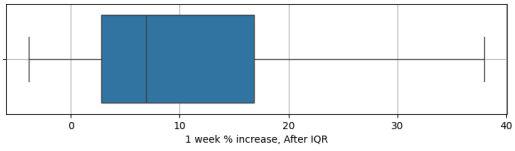

0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
1e6
Confirmed last week, Before IQR


0
20000
40000
60000
80000
Confirmed last week, After IQR


0
100000
200000
300000
400000
1 week change, Before IQR


0
1000
2000
3000
4000
5000
6000
7000
8000
1 week change, After IQR


0
50
100
150
200
1 week % increase, Before IQR


0
10
20
30
40
1 week % increase, After IQR

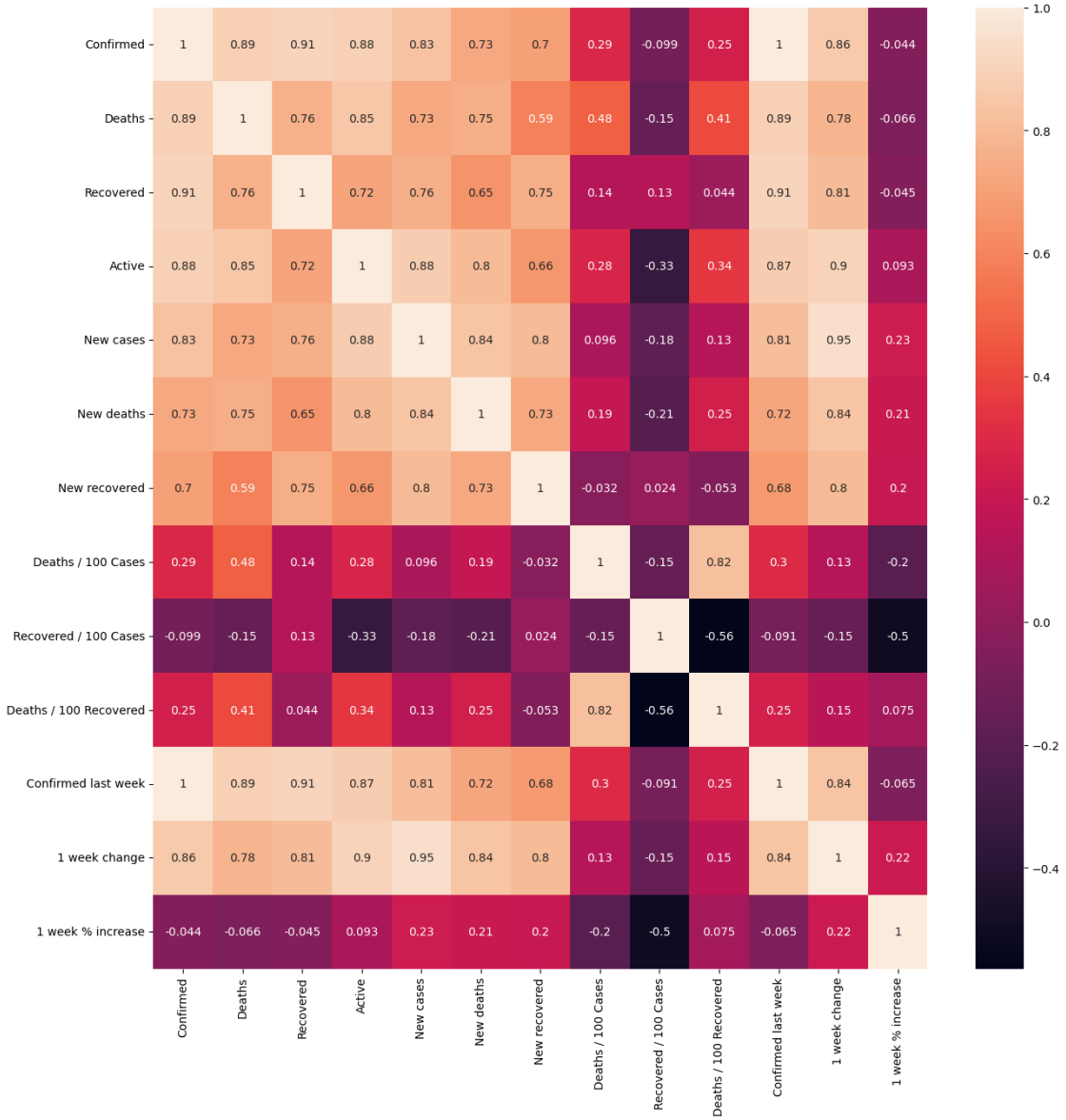
8 Feature Engineering

[17]:
```
#nothing here
```

9 Correlation Matrix

[18]:
```
plt.figure(figsize=(15,15))
sns.heatmap(data[num_features].corr(),annot=True)
```

[18]: <Axes: >

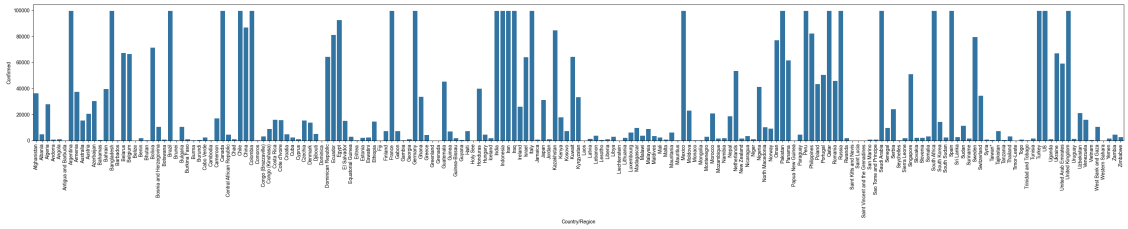


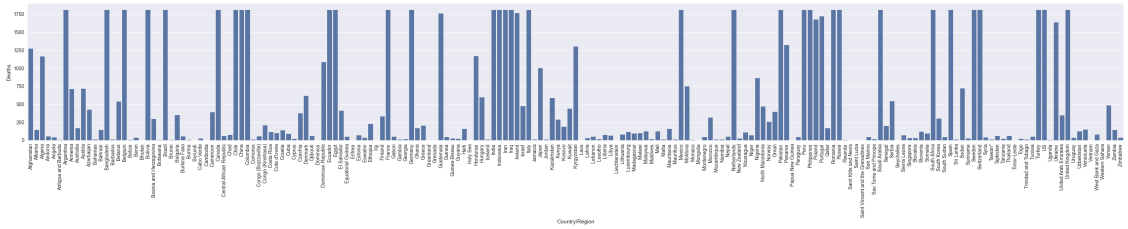
10 Visualization

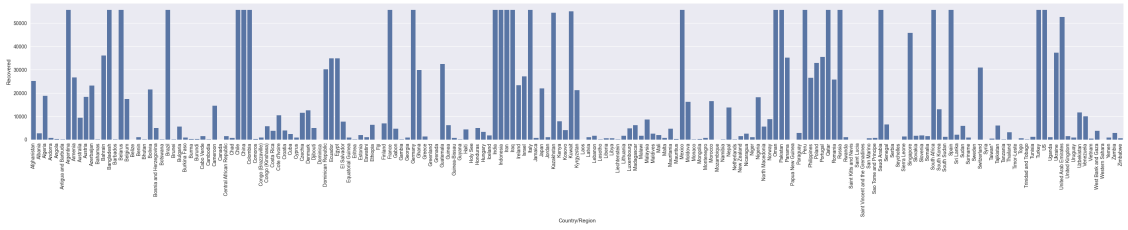
[19]:
```
print(*num_features,sep=', ')
print(*cat_features,sep=', ')
```

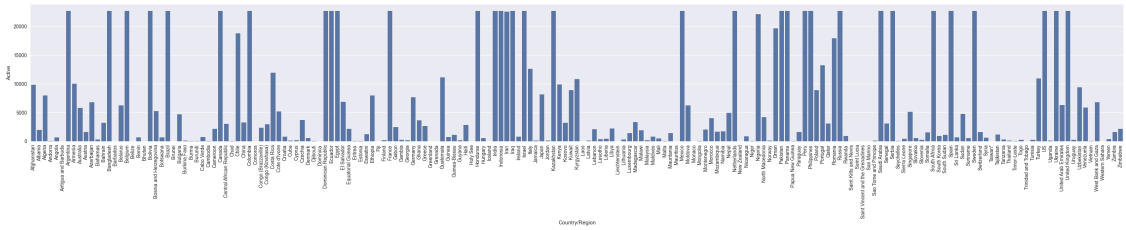
```
Confirmed, Deaths, Recovered, Active, New cases, New deaths, New recovered,
Deaths / 100 Cases, Recovered / 100 Cases, Deaths / 100 Recovered, Confirmed
last week, 1 week change, 1 week % increase
Country/Region, WHO Region
```

[20]:
```
for feature in num_features[0:4]:
    plt.figure(figsize=(40,5))
    plt.xticks(rotation=90)
    sns.barplot(data=data, y=feature, x='Country/Region')
    plt.style.use(plt.style.available[12])
    plt.show()
```









[21]:
```
sns.scatterplot(data=data,x='Deaths',y='Recovered')
```

[21]: <Axes: xlabel='Deaths', ylabel='Recovered'>

